Random Variables

ECON 97: Economic Toolkit

Patrick Molligo

UCLA

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iClicker course link

https://join.iclicker.com/RDNG

Quiz: Conditional Probability

Suppose A and B are independent, with $P(A) > 0$ and $P(B) > 0$. Find:

$$\frac{P(A|B)}{P(A)}$$

A. 0

B. 1

C. It can be determined only if $P(A)$ and $P(B)$ are known

D. It cannot be determined even if $P(A)$ and $P(B)$ are known.

Agenda

1. Recap of material
2. Random Variables

Recap - Conditional Probability & Independence

- $P(A|B) = \frac{P(A \cap B)}{P(B)}$
- If A, B independent ($A \perp B$):
  - $P(A \cap B) = P(A)P(B)$
  - $P(A|B) = P(A)$ and $P(B|A) = P(B)$
  - $A' \perp B'$, $A' \perp B$, and $A \perp B'$
  - $A \perp B \perp C \implies P(A \cap B \cap C) = P(A)P(B)P(C)$

Review

For events A, B: $P(A) = 0.3$, $P(B) = 0.6$, and $P(A \cap B) = 0.2$

What is $P(A \cup B)$?

A. 0.3

B. 0.6

C. 0.7

D. 0.8

Review

For events A, B: $P(A) = 0.3$, $P(B) = 0.6$, and $P(A \cap B) = 0.2$

What is $P(A' \cup B')$?

A. 0.3

B. 0.6

C. 0.7

D. 0.8

Review

For events A, B: $P(A) = 0.3$, $P(B) = 0.6$, and $P(A \cap B) = 0.2$

What is $P(B|A)$?

A. 1/3

B. 2/3

C. 1

D. None of the above

Review

For events A, B: $P(A) = 0.3$, $P(B) = 0.6$, and $P(A \cup B) = 0.8$

What is $P(A' \cup B')$?

A. 0.6

B. 0.7

C. 0.8

D. 0.9

Review

For events A, B: $P(A) = 0.3$, $P(B) = 0.6$, and A, B mutually exclusive

What is $P(A \cup B)$?

A. 0.6

B. 0.7

C. 0.8

D. 0.9

For events A, B: $P(A) = 0.3$, $P(B) = 0.6$, and A, B independent

What is $P(A \cup B)$?

A. 0.64

B. 0.72

C. 0.80

D. 0.90

Review

For events A, B: $P(A) = 0.2$, $P(B) = 0.3$

Which of the following can NOT be a possible value of $P(A \cup B)$?

A. 0.2

B. 0.3

C. 0.4

D. 0.5

Practice - Conditional Probability

Suppose that there are 14 songs on an album and you like 8 of them. When shuffling the album each of the 14 songs is played once in a random order. Find the probability that among the first 2 songs that are played:

1. You like both of them
2. You like neither of them
3. You like exactly one of them

Practice - Independence

A power utility can supply electricity to a city from 3 different power plants. Suppose that each power plant fails with probability 0.2, independent of the others. Suppose that any one plant can provide enough electricity to supply the entire city. What is the probability that the city will experience a black-out?

Now suppose that it takes two plants to provide enough electricity to supply the entire city. What is the probability that the city will experience a black-out?

Practice - Independence

Die A has orange on one face and blue on 5 faces. Die B has orange on 2 faces and blue on 4 faces. Die C has orange on 3 faces and blue on 3 faces. These are unbiased dice. If all of the dice are rolled, find the probability that exactly 2 of the 3 dice come up orange.

Random Variables

Think of a variable X that represents the outcome of an experiment.

The outcome space is called the **support** of X

$\longrightarrow$ E.g., if X is the result of rolling a die, then the support is:

$$S = \{1, 2, 3, 4, 5, 6\}$$

Compute the probability of each outcome using a function called the **PMF**:

$$P(X = x) = f(x)$$

Here, $f(x) = 1/6$ for all values in the support.

The probabilities must add up to 1:

$$\sum_{x \in \{1,\ldots,6\}} f(x) = 1$$

Two functions: PMF vs. CDF

PMF (probability mass function)

$$P(X = x) = f(x)$$

E.g., $f(3) = P(X = 3)$

The PMF must give a probability for every possible value of X

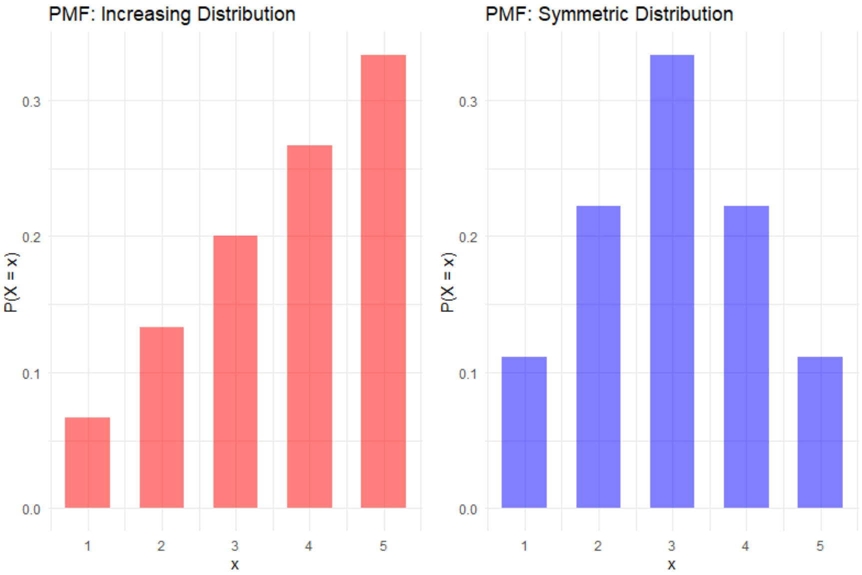
Sum of all values of the PMF is 1 $\longrightarrow \sum_x f(x) = 1$

CDF (cumulative distribution function)

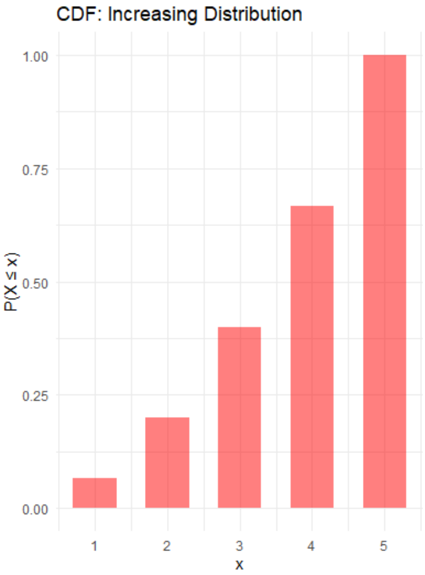
$$P(X \leq x) = F(x)$$

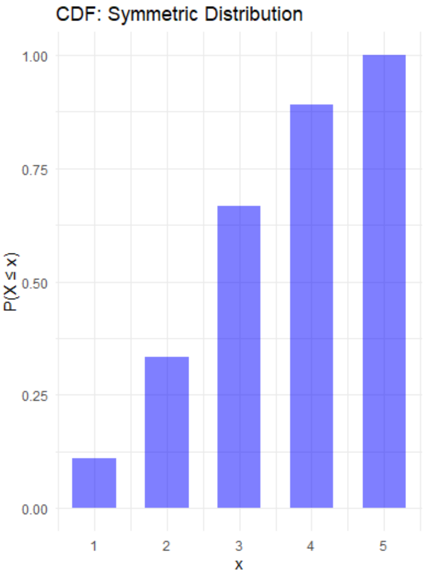
E.g., $F(3) = P(X \leq 3) = f(0) + f(1) + f(2) + f(3)$

PMF plots (histograms)



CDF plots


CDF: Increasing Distribution
P(X ≤ x)
1.00
0.75
0.50
0.25
0.00
1
2
3
4
5
x


CDF: Symmetric Distribution
P(X ≤ x)
1.00
0.75
0.50
0.25
0.00
1
2
3
4
5
x